# Day 01 | Thursday

01

October 29, 2026

## Whole numbers and estimation

9:00 AM–10:15 AM • Math • 75 min

## Calculator basics and checking

10:30 AM–12:00 PM • Calculation Practice • 90 min

**Week 1 • 2 scheduled blocks • 165 teaching minutes**

Estimate. Show the method. Label units. Check the result.

# Today’s exact schedule

## 9:00 AM–10:15 AM • 75 min

### Whole numbers and estimation

| | | |
|---|---|---|
| 9:00 AM–9:05 AM | **Start** | 5 |
| 9:05 AM–9:10 AM | **Recall** | 5 |
| 9:10 AM–9:25 AM | **Teach** | 15 |
| 9:25 AM–9:40 AM | **Together** | 15 |
| 9:40 AM–10:00 AM | **Practise** | 20 |
| 10:00 AM–10:10 AM | **Check** | 10 |
| 10:10 AM–10:15 AM | **Wrap up** | 5 |

Math • Math Week 1, Lesson 1

## 10:30 AM–12:00 PM • 90 min

### Calculator basics and checking

| | | |
|---|---|---|
| 10:30 AM–10:35 AM | **Start** | 5 |
| 10:35 AM–10:40 AM | **Recall** | 5 |
| 10:40 AM–10:55 AM | **Teach** | 15 |
| 10:55 AM–11:10 AM | **Together** | 15 |
| 11:10 AM–11:45 AM | **Practise** | 35 |
| 11:45 AM–11:55 AM | **Check** | 10 |
| 11:55 AM–12:00 PM | **Wrap up** | 5 |

Calculation Practice • Math Week 1, Lesson 2

**Break: 10:15 AM–10:30 AM • 15 minutes • excluded from teaching time**

# Whole numbers and estimation

**Today’s goal: Read whole-number quantities; estimate and calculate a cut-list total; explain why an answer is reasonable.**

Name one use for today’s skill.

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

Exercise Book 1A Q1–10 (p.6), Q11–17 (pp.6–7)

Keep your notebook, textbook, exercise book and chosen calculator ready.

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

# Recall and reconnect

Use Exercise Book 1A Q3,5,9 without a calculator. Discuss which operation finds an offcut.

**Try first without displayed answers.**

Resources: Exercise Book 1A Q1–10 (p.6), Q11–17 (pp.6–7)

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

# Recall from the Exercise Book | 1A Q3

What is the 7 worth in 37,215?

| Ten thousands | Thousands | Hundreds | Tens | Ones |
|---|---|---|---|---|
| 3 | 7 | 2 | 1 | 5 |

Write the operation, working and units before checking.

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

# Read the value before calculating

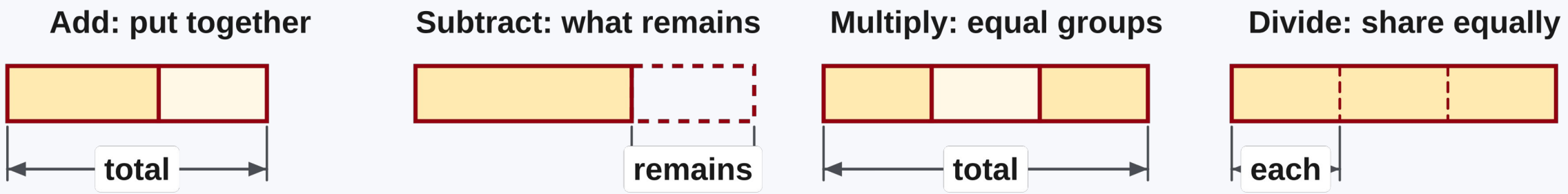
1. A digit’s place tells its value: in 4,072, the 4 means four thousand.
2. Read the quantity and unit together; 1,200 mm is a length.
3. When rounding, name the target place before changing any digits.
4. An estimate uses simpler nearby values; it predicts the size of the answer.

| Thousands | Hundreds | Tens | Ones |
|---|---|---|---|
| **4** | **0** | **7** | **2** |
| 4,000 | 0 | 70 | 2 |

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

# Choose an operation that matches the quantity

1. Add separate quantities to find a total.
2. Subtract the amount used from the starting amount to find what remains.
3. Multiply equal groups; divide to share equally or count groups.
4. Compare the exact result with the estimate and explain any large difference.




Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

# An estimate should agree with the scale

Classroom example: 48 packs contain 21 fasteners each.

**Estimate: 50 × 20 = 1,000 fasteners**
**Calculate: 48 × 21 = 1,008 fasteners**

The exact answer is close to 1,000—not 100 or 10,000.
A calculator result still needs a quantity, unit and reason.




## Work together

Exercise Book 1A Q11, 13 and 15.
Estimate first; alternate solver and checker. Compare hand and calculator methods.

W1 L1 • Selected calculator: TI-30Xa or Sharp EL-531XT

# Textbook | page 14

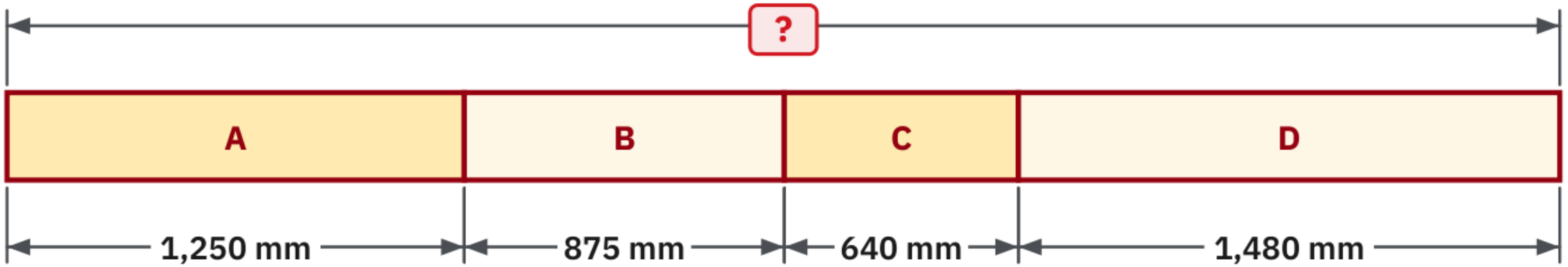



**Figure 1.2** The cut list from Example 1.4. The total length is hidden.

Textbook p.14 · Figure 1.2: read the cut-list dimensions.

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

# Textbook | page 14

## Example 1.4 Estimating a cut list

A cut list has four pieces: 1,250 mm, 875 mm, 640 mm, 1,480 mm. Estimate the total length.

**SOLUTION**

1. Round each length to the nearest hundred: 1,300, 900, 600, 1,500.
2. Add the rounded numbers: 1,300 + 900 + 600 + 1,500 = 4,300.

**Answer** **About 4,300 mm. The exact total, found in Example 1.5, is 4,245 mm, which is close to the estimate.**

Textbook p.14 · Example 1.4: estimate before calculating.

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

# Spot the mistake. Explain the correction.

## A learner estimates 48 × 21 as 1,000, then accepts a calculator result of 100. What check failed?

Discuss the method before checking the answer.

**The result is about ten times too small. Re-enter the intended multiplication and compare again.**

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

# Work together

Work Exercise Book 1A Q11,13,15 together. Estimate first, then compare hand and calculator methods.

**Work in pairs; alternate solver and checker.**

Resources: Exercise Book 1A Q1–10 (p.6), Q11–17 (pp.6–7)

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

# Guided practice | 1A Q11

3,275 + 1,948

$$\begin{array}{r} 3{,}275 \\ +\ 1{,}948 \\ \hline \end{array}$$

Work with a partner. Take turns explaining the next step.

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

# Compare your method | 1A Q11

3,275 + 1,948

**CHECK TOGETHER**

**5,223**

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

# Your scheduled practice

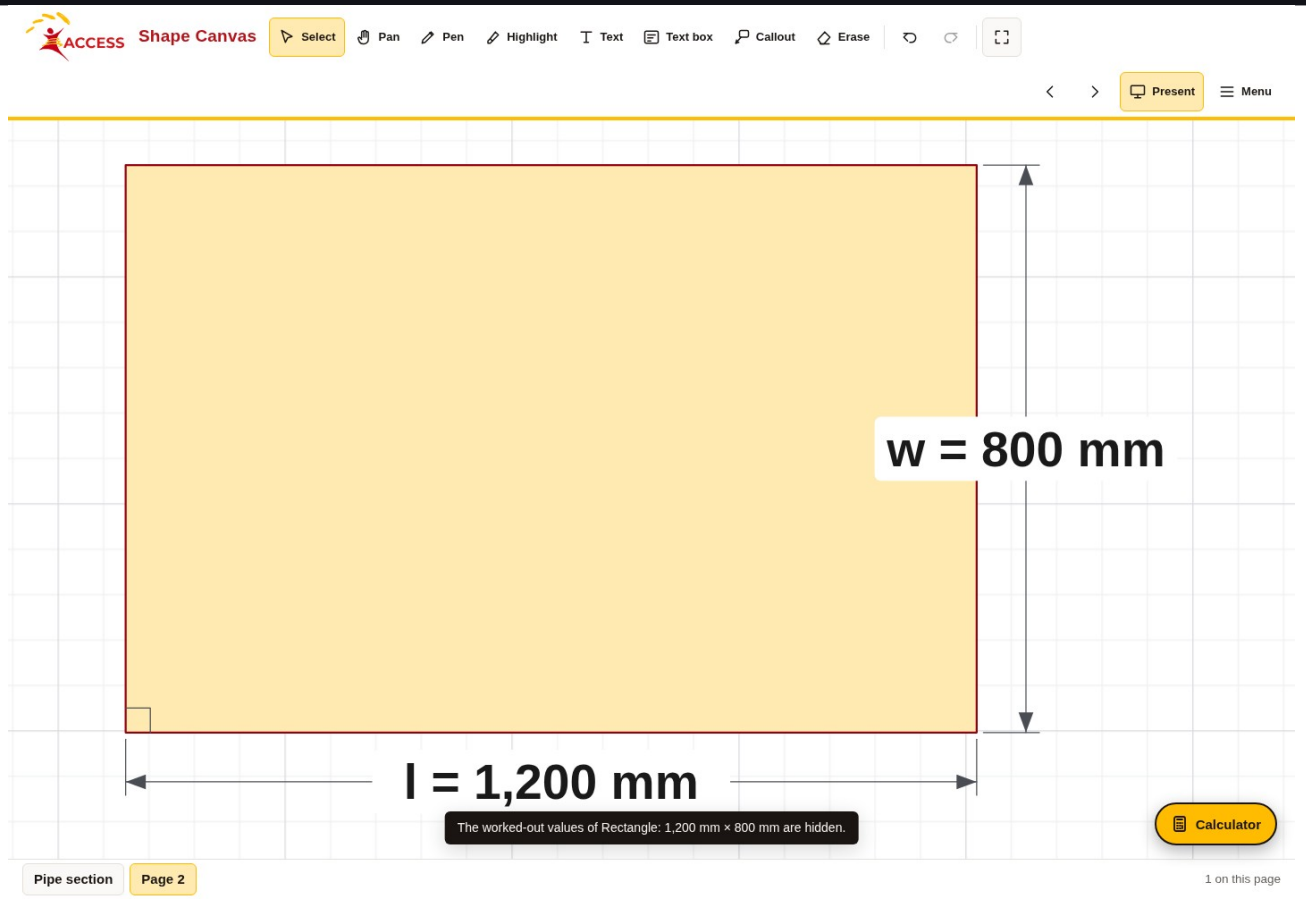
Complete Exercise Book 1A Q1–10 and Q12,14,16,17 as time allows. On Shape Canvas, set a rectangle to 1,200 mm by 800 mm and read both sizes only, since area and perimeter come later in the course.

**Show working and units; estimate, then check.**

Resources: Exercise Book 1A Q1–10 (p.6), Q11–17 (pp.6–7)

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

# Open Shape Canvas




Navigation example. Use the values and units in the assigned task.

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

# Practise in Numeracy Hub and Shape Canvas

1. Supported: In Numeracy Hub Calculator, select TI-30Xa or Sharp EL-531XT and clear the display.

2. Independent practice: Estimate and write your assigned 1A calculation before checking its result.

3. Supported preview: In Shape Canvas, label a 1,200 mm × 800 mm rectangle; read both lengths only.

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

# Practise from the Exercise Book | 1A Q1

Write 4,072 in words.

| Thousands | Hundreds | Tens | Ones |
|---|---|---|---|
| 4 | 0 | 7 | 2 |

Write the operation, working and units before checking.

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

# Practise from the Exercise Book | 1A Q2

Write in digits: fifteen thousand three hundred eight.

| Ten thousands | Thousands | Hundreds | Tens | Ones |
|---|---|---|---|---|
| | | | | |

Write the operation, working and units before checking.

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

# Check independently

Exercise Book 1A Q5 and Q10; explain one estimate and show the operation used.

**Answer independently and show the method.**

Resources: Exercise Book 1A Q1–10 (p.6), Q11–17 (pp.6–7)

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

# Independent check | 1A Q5

Round 3,462 to the nearest ten.

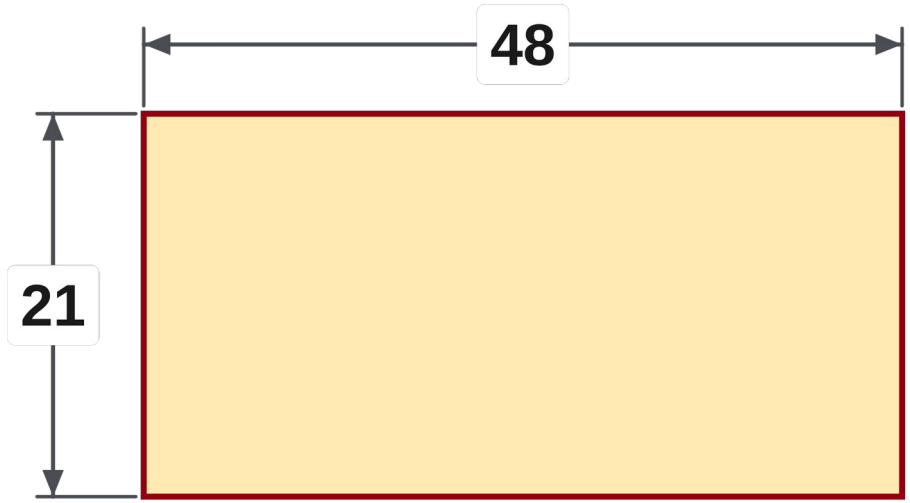
| Thousands | Hundreds | Tens | Ones |
|---|---|---|---|
| 3 | 4 | 6 | 2 |

Answer independently. Keep worked solutions closed.

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

# Independent check | 1A Q10

Estimate 48 × 21 by rounding each number to the nearest ten.




Answer independently. Keep worked solutions closed.

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

# Wrap up and choose the next step

Share one learning point and one skill to practise next.

**Record a takeaway and next action.**

Resources: Exercise Book 1A Q1–10 (p.6), Q11–17 (pp.6–7)

Textbook Chapter 1 pp.12–23; Chapter 2 pp.48–51,59

# Break

## 10:15 AM–10:30 AM

Return ready for: Calculator basics and checking

**This break is excluded from teaching minutes.**

# Calculator basics and checking

**Today’s goal: Use the chosen calculator correctly and check an answer with an estimate or inverse operation.**

Name one use for today’s skill.

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

Exercise Book 2A Q1–3,6 (p.17); 1A Q11–17,20 (pp.6–7)

Keep your notebook, textbook, exercise book and chosen calculator ready.

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

# Recall and reconnect

Try to clear the display and enter 23 × 23. Check which model they use.

**Try first without displayed answers.**

Resources: Exercise Book 2A Q1–3,6 (p.17); 1A Q11–17,20 (pp.6–7)

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

# Know what the key will do

1. Clear the old calculation before beginning a new problem.

2. Use the correction key to repair an entry; reread the intended calculation.

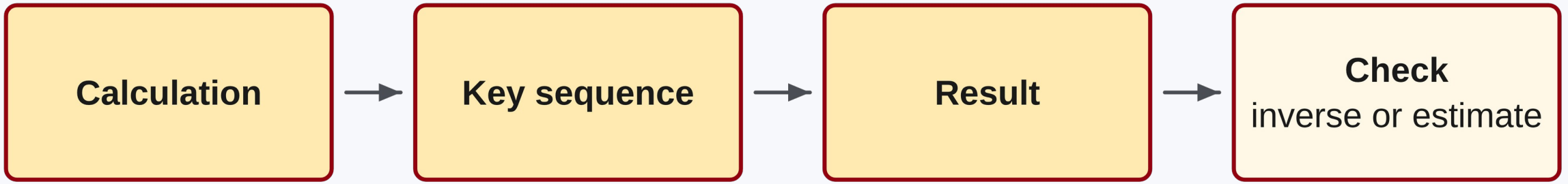
3. The square key multiplies a number by itself; square root reverses this for positive values.

4. Change sign reverses a value’s sign; subtraction finds the difference between two values.

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

# Keep a record that can be checked

1. Write calculation → key sequence → result, using your selected calculator model.
2. For TI-30Xa and Sharp, follow the textbook's own square-root entry sequence.
3. Use an inverse operation: multiply a quotient back or square a square-root result.
4. Use an estimate as a separate check that the answer has the expected size.

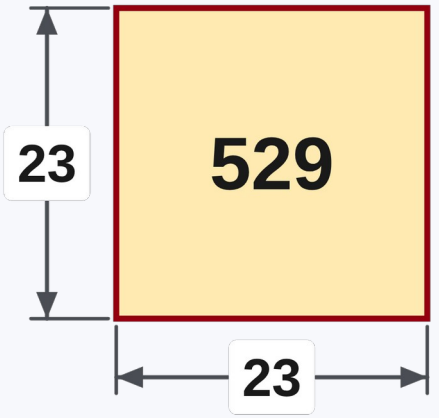



Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

# Reverse the operation to check

$23^2 = 23 \times 23 = 529$
$\sqrt{529} = 23$

Squaring and square root undo one another for these positive values.
Write the intended calculation before entering keys.




Square
**$23^2 = 529$**

Square root
**$\sqrt{529} = 23$**

## A second classroom example

2,688 ÷ 12 = 224 pieces
Check: 224 × 12 = 2,688. An estimate near 225 supports the result.

## Work together

Exercise Book 2A Q1–3, p.17. Record calculation, keys and answer; compare root entry on the two models.

# Textbook | page 49

| Job | TI-30Xa | Sharp EL-531XT |
|---|---|---|
| Turn on | ON/C | ON/C |
| Turn off | OFF | 2ndF then ON/C (OFF is printed above it) |
| Second functions | 2nd | 2ndF |
| Change the sign of a number | +⇄− after the digits | +/− before the number |
| Square root | √x after the number | √ before the number, then = |
| Square | $x^2$ after the number | $x^2$ after the number, then = |
| Store in memory | STO then 1, 2 or 3 | STO then a letter key (A to F, X, Y, M) |
| Fix a mistyped digit | ← | DEL |

Textbook p.49 · Table 2.1: key differences between the two calculators.

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

# Textbook | page 52

## CALCULATOR

### TI-30Xa

$12^2$

[12] [x²]

Display 144.

√144

[144] [√x]

Display 12.

### Sharp EL-531XT

$12^2$

[12] [x²] [=]

Display 144.

√144

[√] [144] [=]

Display 12.

**Answer** $12^2 = 144$ and $\sqrt{144} = 12$. Each answer checks the other.

Textbook p.52 · Square and square-root key sequences; each checks the other.

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

# Spot the mistake. Explain the correction.

## Does pressing the subtraction key necessarily enter a negative number correctly?

Discuss the method before checking the answer.

**No. Use the selected model’s change-sign key for a negative value; use subtraction between values.**

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

# Work together

Solve Exercise Book 2A questions 1–3 together. Compare square-root entry on the two models; estimate before checking.

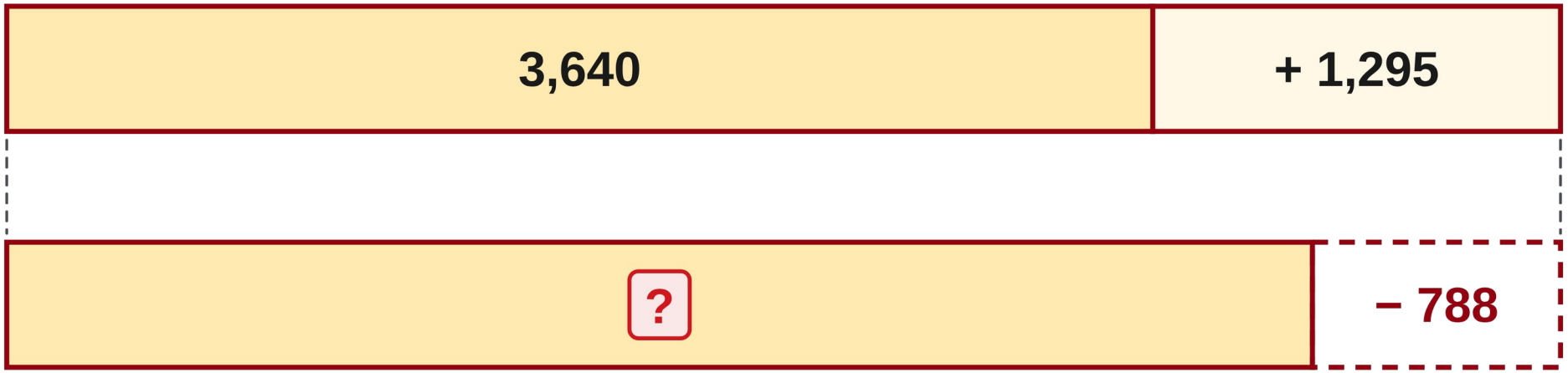
**Work in pairs; alternate solver and checker.**

Resources: Exercise Book 2A Q1–3,6 (p.17); 1A Q11–17,20 (pp.6–7)

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

# Guided practice | 2A Q1

3,640 + 1,295 − 788




Work with a partner. Take turns explaining the next step.

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

# Compare your method | 2A Q1

3,640 + 1,295 − 788

**CHECK TOGETHER**

**4,147. TI-30Xa: 3640, +, 1295, −, 788, =. Sharp: 3640, +, 1295, −, 788, =.**

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

# Your scheduled practice

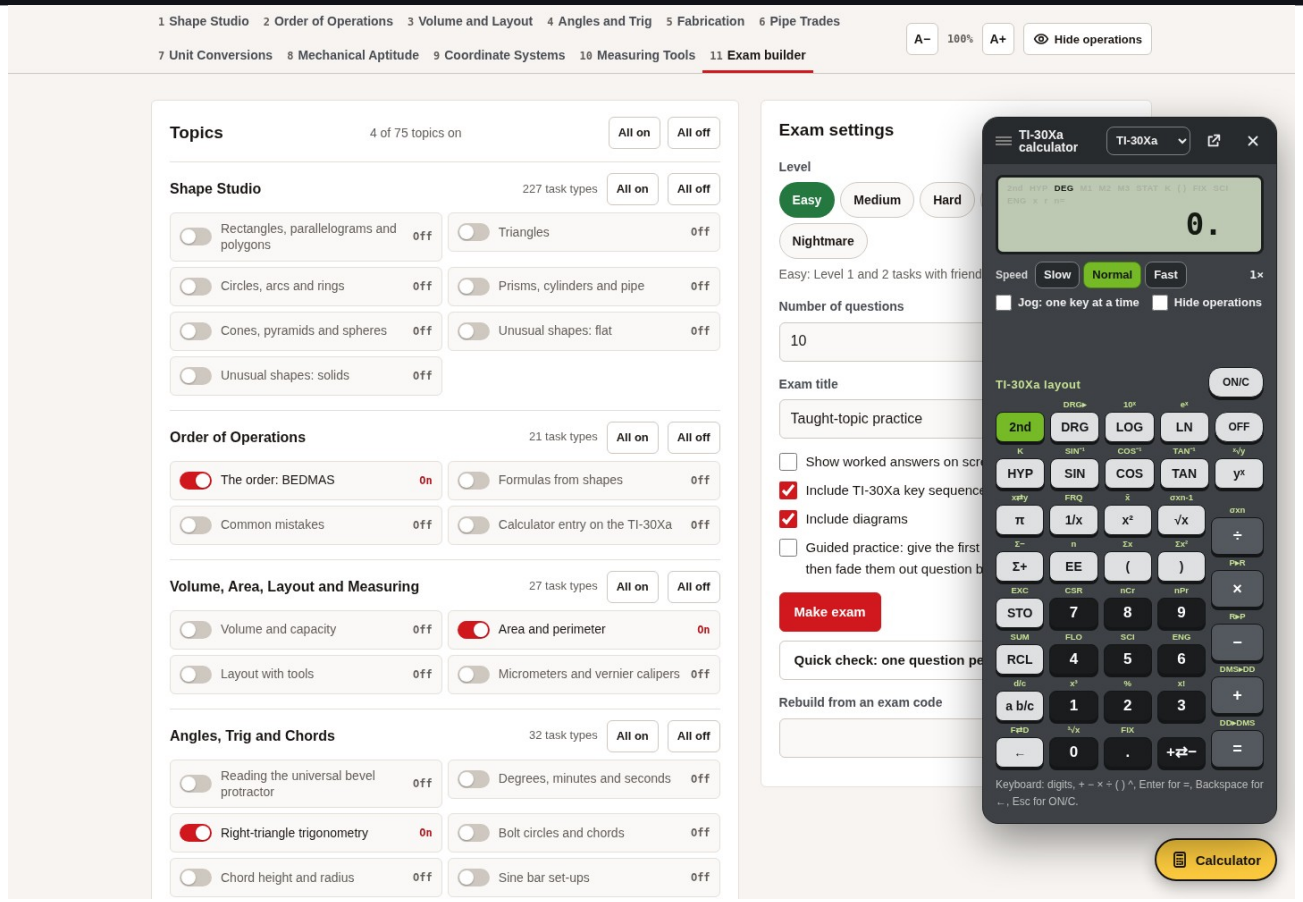
Complete 2A question 6 and selected 1A questions 11–17,20. Record the calculation, keys and answer; use inverse operations to verify two results.

**Show working and units; estimate, then check.**

Resources: Exercise Book 2A Q1–3,6 (p.17); 1A Q11–17,20 (pp.6–7)

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

# Open Numeracy Hub




Navigation example. Use the values and units in the assigned task.

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

# Practise in Numeracy Hub

1. Supported: Choose one calculator model in Numeracy Hub; locate clear, square, root and sign keys.

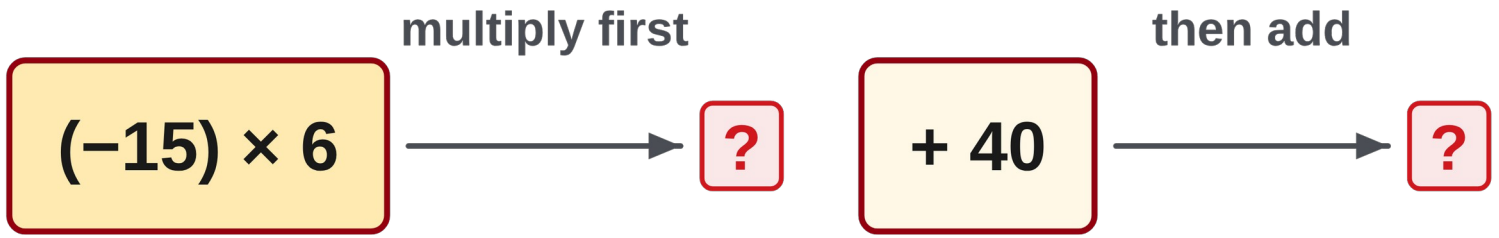
2. Supported: Work 2A Q1–3 and record each model-specific key sequence beside the written calculation.

3. Independent practice: Check two completed calculations using inverse operations; retain the working.

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

# Practise from the Exercise Book | 2A Q6

$(-15) \times 6 + 40$




Write the operation, working and units before checking.

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

# Practise from the Exercise Book | 1A Q11

3,275 + 1,948

$$\begin{array}{r} 3{,}275 \\ +\ 1{,}948 \\ \hline \end{array}$$

Write the operation, working and units before checking.

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

# Check independently

Redo Exercise Book 2A questions 1 and 3 independently. Explain one checking method.

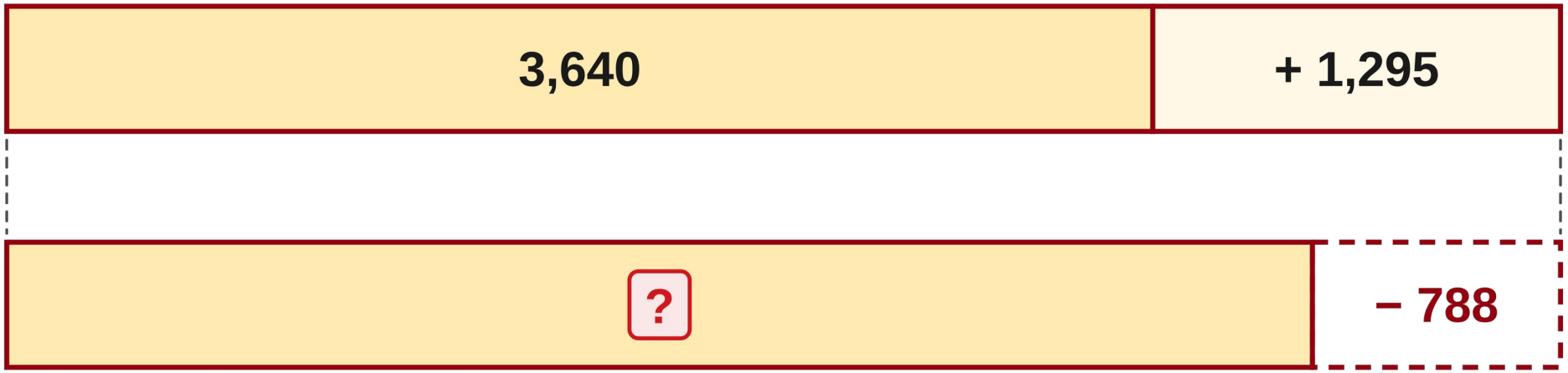
**Answer independently and show the method.**

Resources: Exercise Book 2A Q1–3,6 (p.17); 1A Q11–17,20 (pp.6–7)

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

# Independent check | 2A Q1

3,640 + 1,295 − 788

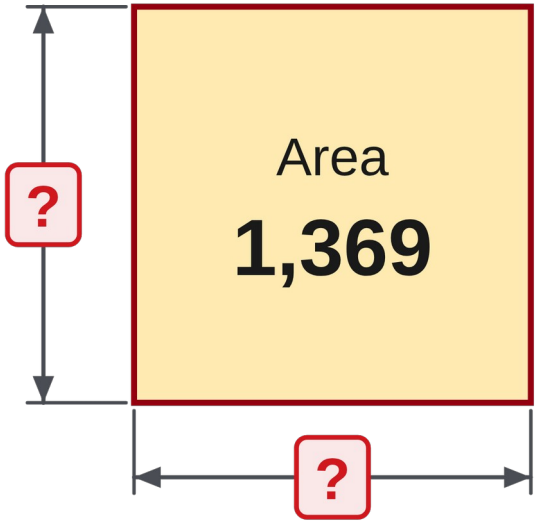



Answer independently. Keep worked solutions closed.

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

# Independent check | 2A Q3

√1,369




Answer independently. Keep worked solutions closed.

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23

# Wrap up and choose the next step

Share one learning point and one skill to practise next.

**Record a takeaway and next action.**

Resources: Exercise Book 2A Q1–3,6 (p.17); 1A Q11–17,20 (pp.6–7)

Textbook Chapter 2 pp.48–53,59; Chapter 1 pp.14–23